1	**(a)**		The principle of conservation of linear momentum states that the total momentum of a system of interacting bodies <u>remain constant</u> provided <u>no external force</u> acts on the system.	1 1
	(b)	**(i)**	By the principle of conservation of momentum, $m_X u_X + m_Y u_Y = m_X v_X + m_Y v_Y$ 4.0(500) + 28.0(340) = 4.0(220) + 28.0(v) v = 380 m s^{-1}	 1 1
		(ii)	Relative speed of approach = $u_X - u_Y$ = 500 – 340 = 160 m s^{-1} Relative speed of separation = $v_Y - v_X$ = 380 – 220 = 160 m s^{-1} Since the relative speeds of approach and separation are the same, the collision is elastic. Or Check whether total final kinetic energy equals the total initial kinetic energy.	1 1
		(iii)	The forces on the two bodies (or on X and on Y) are equal in magnitude and opposite in direction The duration of impact for both forces is the same <u>and</u> force is change in momentum / time	1 1

2	(a)	On a banked track, the normal contact force by the road on the car will be tilted	